

IOANNIS ILIADIS

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TECHNICAL SKILLS

Languages: C/C++ (Advanced), Python, Java, C#, PHP

APIs: Vulkan, OpenGL, DirectX 12, Windows API

Developer Tools: Git/GitHub, CMake, CLion, RenderDoc, Jira, VS2022

Software Engineering: OOP, Data Structures & Algorithms, Concurrency

STEM: Linear Algebra, Calculus, Vector Calculus, Mechanics

EDUCATION

Swinburne University of Technology

Bachelor of Computer Science

GPA: 3.6/4.0

Hawthorn, Victoria

Aug. 2022 – July 2025

Melbourne Polytechnic

Advanced Diploma of Engineering

Preston, Victoria

Feb. 2020 – Dec. 2021

PROJECTS

Vulkan Rendering Engine | C++, Vulkan

- Built a physically based forward+ rendering engine using the Vulkan API.
- Implemented **advanced rendering techniques**, such as image based lighting, bloom, SSAO, shadow mapping, skybox, MSAA, and HDR.
- Boosted rendering performance by incorporating **optimizations techniques**, such as instanced rendering, compute-based light culling, async loading, and face culling.
- Developed a **custom editor** with a functional scene graph, infinite grid, transformation gizmos, world-space light icons, drag & drop, and popups.
- **Preview:** John-Iliadis/VulkanRenderingEngine

Projectile Motion Simulator | C++, OpenGL

- Developed a **physics-based simulator** for modeling motion in two dimensions.
- Implemented a 2D batch renderer to **optimize draw calls**.
- Implemented **custom text rendering** using FreeType for font rasterization.
- Created interactive vector, graph, and trajectory **visualizations** for analyzing the projectile's motion.
- **Preview:** John-Iliadis/ProjectileMotionSimulation

Pixel Adventure | C++, SFML

- A **fast paced 2D platformer** developed with SFML as the graphics library.
- Implemented **core systems**, including physics, animations, GUI, and asset handling.
- Parsed and managed world and configuration data with the nlohmann::json library.
- Used the Tiled editor for map design and creation.
- **Preview:** John-Iliadis/PixelAdventure

WORK EXPERIENCE

Warehouse Team Member

Archeiothiki SA

Managed daily warehouse operations including scanning, order picking with lifts, stocking shelves, and palletizing using electric equipment.

March 2022 – Aug. 2022

Athens, Greece

Trades Assistant

Iliadis Renovations

Assisted with construction tasks including demolition, material handling, tool operation, and site clean-up.

Feb. 2020 – Dec. 2021

Melbourne, Australia

CERTIFICATIONS

AWS Certified Cloud Practitioner

Dec. 2023